

Applied Statistics for Engineering and Scientists II

Rewritten Questions, Complete Solutions, and Casio fx-580VN X Checks

Question 1: Multiple Linear Regression (4 points)

A manufacturing engineer studies the relationship between daily machine output y , operating time x_1 (hours), and technician support time x_2 (hours). The observations from six production days are

Day	1	2	3	4	5	6
x_1	1	2	3	4	5	6
x_2	3	1	2	2	1	3
y	9.5	6.0	8.0	10.0	9.5	11.0

Consider the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

- (a) Using the centered normal equations, compute the least-squares estimates $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$. Then compute R^2 .
- (b) The reduced model omits x_2 . The nested-model comparison in R is

Model	Res.Df	RSS	Df	Sum of Sq	F	$\text{Pr}(> F)$
$y \sim x_1$	4	17.500				
$y \sim x_1 + x_2$	3	1.500	1	16.000	32.000	0.01094

State the null and alternative hypotheses. At level $\alpha = 0.05$, determine whether the full model gives a statistically significant improvement.

- (c) Predict daily output when $x_1 = 4.5$ and $x_2 = 2.5$.

Solution

The sample means are

$$\bar{x}_1 = \frac{1+2+3+4+5+6}{6} = 3.5, \quad \bar{x}_2 = \frac{3+1+2+2+1+3}{6} = 2, \quad \bar{y} = 9.$$

Define centered variables

$$u_i = x_{i1} - \bar{x}_1, \quad v_i = x_{i2} - \bar{x}_2, \quad w_i = y_i - \bar{y}.$$

The required centered sums are

$$S_{11} = \sum u_i^2 = 17.5, \quad S_{22} = \sum v_i^2 = 4, \quad S_{12} = \sum u_i v_i = 0,$$

$$S_{1y} = \sum u_i w_i = 10, \quad S_{2y} = \sum v_i w_i = 5.$$

Therefore, the centered normal equations are

$$\begin{bmatrix} 17.5 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}.$$

Hence

$$\hat{\beta}_1 = \frac{10}{17.5} = \frac{4}{7}, \quad \hat{\beta}_2 = \frac{5}{4} = 1.25.$$

The intercept is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2 = 9 - \frac{4}{7}(3.5) - \frac{5}{4}(2) = 4.5.$$

Thus the fitted regression equation is

$$\hat{y} = 4.5 + \frac{4}{7}x_1 + 1.25x_2.$$

The fitted values are

$$\hat{y} = \left(\frac{247}{28}, \frac{193}{28}, \frac{244}{28}, \frac{260}{28}, \frac{241}{28}, \frac{327}{28} \right),$$

and the residual sum of squares is

$$\text{RSS} = \sum_{i=1}^6 (y_i - \hat{y}_i)^2 = \frac{99}{28} \approx 3.5357.$$

Also,

$$\text{SST} = \sum_{i=1}^6 (y_i - \bar{y})^2 = 15.5.$$

Therefore,

$$R^2 = 1 - \frac{\text{RSS}}{\text{SST}} = 1 - \frac{3.5357}{15.5} \approx 0.7719.$$

Hence

$$R^2 \approx 0.772.$$

Approximately 77.2% of the variation in daily output is explained by x_1 and x_2 .
For the nested-model test,

$$H_0 : \beta_2 = 0 \quad \text{versus} \quad H_a : \beta_2 \neq 0.$$

Since

$$p = 0.01094 < 0.05,$$

we reject H_0 . Thus, adding technician support time x_2 produces a statistically significant improvement over the reduced model.

For $x_1 = 4.5$ and $x_2 = 2.5$,

$$\hat{y} = 4.5 + \frac{4}{7}(4.5) + 1.25(2.5) = 10.1964.$$

Therefore,

$$\hat{y} \approx 10.20.$$

Question 2: Power-Law Transformation (3 points)

In a laboratory experiment, a positive response Y is modeled as a power law in a positive input X :

$$Y = \alpha X^\beta \exp(\varepsilon), \quad \varepsilon \sim N(0, \sigma^2), \quad \alpha > 0.$$

The observed data are

x	1	2	4	8
y	3	12	48	192

Here and below, $\log$ denotes the natural logarithm.

- Transform the model into a linear regression model. Using the transformed data, compute the least-squares estimates $\hat{\alpha}$ and $\hat{\beta}$.
- Using the plug-in fitted curve $\hat{y}(x) = \hat{\alpha}x^{\hat{\beta}}$, predict the response at $x = 16$. No lognormal bias correction is requested.

Solution

Take natural logarithms:

$$\log Y = \log \alpha + \beta \log X + \varepsilon.$$

Set

$$Z = \log Y, \quad W = \log X, \quad \gamma_0 = \log \alpha.$$

Then

$$Z = \gamma_0 + \beta W + \varepsilon,$$

which is a simple linear regression model.

The transformed observations are

W	0	$\log 2$	$2 \log 2$	$3 \log 2$
Z	$\log 3$	$\log 3 + 2 \log 2$	$\log 3 + 4 \log 2$	$\log 3 + 6 \log 2$

so every observation lies exactly on

$$Z = \log 3 + 2W.$$

Therefore,

$$\hat{\gamma}_0 = \log 3, \quad \hat{\beta} = 2.$$

Since $\gamma_0 = \log \alpha$,

$$\hat{\alpha} = e^{\hat{\gamma}_0} = e^{\log 3} = 3.$$

Thus

$$\boxed{\hat{\alpha} = 3, \quad \hat{\beta} = 2.}$$

The fitted curve is

$$\hat{y}(x) = 3x^2.$$

At $x = 16$,

$$\hat{y}(16) = 3(16)^2 = 3(256) = 768.$$

Hence

$$\boxed{\hat{y}(16) = 768.}$$

Question 3: Logistic Regression (2 points)

Let $Y = 1$ indicate machine failure during the next operating cycle, and let $Y = 0$ indicate no failure. The predictors are temperature x_1 , vibration level x_2 , and a maintenance indicator x_3 , where $x_3 = 1$ if preventive maintenance was performed and $x_3 = 0$ otherwise.

An R fit gives

Coefficient	Estimate	Std. Error	z value	Pr(> z)
Intercept	-4.200	1.300	-3.231	0.00124
x_1	0.060	0.020	3.000	0.00270
x_2	1.100	0.300	3.667	0.00025
x_3	-1.500	0.550	-2.727	0.00639

- (a) Write the fitted logistic regression model and interpret the three slope coefficients in terms of odds, holding the other predictors fixed.
- (b) Classify the following observations using threshold 0.5. You may compare the fitted log-odds with zero instead of evaluating the logistic function numerically.

Observation	x_1	x_2	x_3
1	50	1.0	1
2	60	2.0	0
3	55	1.5	1
4	70	0.5	0

Solution

Let

$$p(x) = P(Y = 1 \mid x_1, x_2, x_3).$$

The fitted logit model is

$$\log\left(\frac{\hat{p}}{1 - \hat{p}}\right) = -4.2 + 0.06x_1 + 1.1x_2 - 1.5x_3.$$

Equivalently,

$$\hat{p} = \frac{\exp(-4.2 + 0.06x_1 + 1.1x_2 - 1.5x_3)}{1 + \exp(-4.2 + 0.06x_1 + 1.1x_2 - 1.5x_3)}.$$

Holding all other predictors fixed:

- A one-unit increase in temperature multiplies the odds of failure by

$$e^{0.06} \approx 1.0618.$$

Thus, the odds increase by about 6.18%.

- A one-unit increase in vibration level multiplies the odds by

$$e^{1.1} \approx 3.004.$$

Thus, the odds are approximately tripled.

- Preventive maintenance changes x_3 from 0 to 1 and multiplies the odds by

$$e^{-1.5} \approx 0.2231.$$

Thus, maintenance reduces the odds of failure by about

$$1 - 0.2231 = 0.7769,$$

or 77.69%.

Using a threshold of 0.5 is equivalent to predicting failure when the fitted log-odds

$$\hat{\eta} = -4.2 + 0.06x_1 + 1.1x_2 - 1.5x_3$$

is positive.

Observation	Fitted log-odds $\hat{\eta}$	Classification
1	$-4.2 + 0.06(50) + 1.1(1) - 1.5(1) = -1.6$	$\hat{Y} = 0$
2	$-4.2 + 0.06(60) + 1.1(2) - 1.5(0) = 1.6$	$\hat{Y} = 1$
3	$-4.2 + 0.06(55) + 1.1(1.5) - 1.5(1) = -0.75$	$\hat{Y} = 0$
4	$-4.2 + 0.06(70) + 1.1(0.5) - 1.5(0) = 0.55$	$\hat{Y} = 1$

Therefore, the four predicted classes are

$$\boxed{0, 1, 0, 1.}$$

Question 4: Properties of the OLS Estimator (1 point)

Let

$$Y = X\beta + \varepsilon,$$

where $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$ is deterministic with $\text{rank}(X) = p$, $\beta \in \mathbb{R}^p$, $E(\varepsilon) = 0_n$, and

$$\text{Var}(\varepsilon) = \sigma^2 I_n.$$

The ordinary least-squares estimator is

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

No Gaussian assumption is needed.

(a) Show that

$$\hat{\beta} - \beta = (X^T X)^{-1} X^T \varepsilon,$$

and use this identity to prove $E(\hat{\beta}) = \beta$.

(b) Using the same identity, prove

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}.$$

Solution

Substitute $Y = X\beta + \varepsilon$ into the OLS formula:

$$\begin{aligned}\hat{\beta} &= (X^T X)^{-1} X^T Y \\ &= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\ &= (X^T X)^{-1} X^T X\beta + (X^T X)^{-1} X^T \varepsilon \\ &= \beta + (X^T X)^{-1} X^T \varepsilon.\end{aligned}$$

Therefore,

$$\boxed{\hat{\beta} - \beta = (X^T X)^{-1} X^T \varepsilon.}$$

Taking expectations,

$$\begin{aligned}E(\hat{\beta}) &= E[\beta + (X^T X)^{-1} X^T \varepsilon] \\ &= \beta + (X^T X)^{-1} X^T E(\varepsilon) \\ &= \beta + (X^T X)^{-1} X^T 0_n \\ &= \beta.\end{aligned}$$

Thus,

$$\boxed{E(\hat{\beta}) = \beta,}$$

so the OLS estimator is unbiased.

For the variance, let

$$A = (X^T X)^{-1} X^T.$$

Then $\hat{\beta} - \beta = A\varepsilon$, so

$$\begin{aligned}\text{Var}(\hat{\beta}) &= \text{Var}(A\varepsilon) \\ &= A \text{Var}(\varepsilon) A^T \\ &= (X^T X)^{-1} X^T (\sigma^2 I_n) [(X^T X)^{-1} X^T]^T.\end{aligned}$$

Since $X^T X$ is symmetric,

$$[(X^T X)^{-1} X^T]^T = X (X^T X)^{-1}.$$

Hence

$$\begin{aligned}\text{Var}(\hat{\beta}) &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1}.\end{aligned}$$

Therefore,

$$\boxed{\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}.}$$

Casio fx-580VN X: Fast Result-Checking Guide

This section is designed for *checking* arithmetic after you have shown the required statistical method. Button names can appear in English or Vietnamese depending on the calculator language. The general path is

MENU $\rightarrow$ choose a mode $\rightarrow$ =

Use the calculator to verify numerical results, but still write formulas, hypotheses, and interpretations in the exam.

Important limitation

The fx-580VN X performs one-variable statistics and two-variable regressions, but it does not directly fit a multiple regression with both x_1 and x_2 in one STAT command. For Question 1, use either the centered-sum shortcut or the Matrix method described below.

1. Question 1: Multiple regression

Method A: fastest check using the special structure of the data

Because the centered cross-product is $S_{12} = 0$, the two slopes can be checked separately:

$$\hat{\beta}_1 = \frac{S_{1y}}{S_{11}}, \quad \hat{\beta}_2 = \frac{S_{2y}}{S_{22}}.$$

Casio shortcut

Use **Statistics** $\rightarrow$ **1-Variable** three times to check the means:

$$\bar{x}_1 = 3.5, \quad \bar{x}_2 = 2, \quad \bar{y} = 9.$$

Then enter the centered products directly in Calculate mode:

$$10 \div 17.5 = 0.571428\dots, \quad 5 \div 4 = 1.25.$$

Finally check

$$9 - (0.571428\dots)(3.5) - (1.25)(2) = 4.5.$$

Store repeated values with **ST0** if desired, for example store 4/7 in A and 1.25 in B.

Method B: general matrix check

For a model with an intercept,

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

For this data,

$$X^T X = \begin{bmatrix} 6 & 21 & 12 \\ 21 & 91 & 42 \\ 12 & 42 & 28 \end{bmatrix}, \quad X^T y = \begin{bmatrix} 54 \\ 199 \\ 113 \end{bmatrix}.$$

Casio shortcut

Choose **MENU** → **Matrix**.

1. Define MatA as a 3×3 matrix and enter $X^T X$.
2. Define MatB as a 3×1 matrix and enter $X^T y$.
3. On the calculation screen evaluate $\text{MatA}^{-1} \text{MatB}$.

The output should be the coefficient vector $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)^T$.

Expected check

$$\text{MatA}^{-1} \text{MatB} = \begin{bmatrix} 4.5 \\ 0.5714285714 \\ 1.25 \end{bmatrix}.$$

To check the prediction, type

$$4.5 + \frac{4}{7}(4.5) + 1.25(2.5) = 10.19642857.$$

Checking R^2 efficiently

Once the coefficients are known, calculate the six fitted values and residuals. The quickest exam workflow is to type each squared residual and add them with the calculator's answer memory:

$$\text{RSS} = 3.535714286, \quad \text{SST} = 15.5, \quad R^2 = 1 - \text{RSS} / \text{SST} = 0.7718894.$$

A useful alternative is to place the six residuals in **Statistics** → **1-Variable**; then use the sum-of-squares statistic $\sum x^2$.

2. Question 2: transformed power-law regression

Transform the columns to

$$W = \ln x, \quad Z = \ln y.$$

Casio shortcut

Choose **MENU** → **Statistics** → linear regression $y = a + bx$. Enter W in the x column and Z in the y column. Open the regression calculation menu and display a , b , and r .

Expected check

The calculator should return approximately

$$a = 1.098612289 = \ln 3, \quad b = 2, \quad r = 1.$$

Therefore

$$\hat{\alpha} = e^a = 3, \quad \hat{y}(16) = 3(16)^2 = 768.$$

An even faster pattern check is that every time x doubles, y is multiplied by 4, so $2^\beta = 4$ and hence $\beta = 2$.

3. Question 3: logistic regression

The calculator does not fit this logistic model from the coefficient table, because the coefficients are already supplied. It is most useful for evaluating odds ratios, log-odds, and probabilities.

Casio shortcut

For each row, calculate

$$\eta = -4.2 + 0.06x_1 + 1.1x_2 - 1.5x_3.$$

With threshold 0.5, do not compute the full logistic probability unless requested:

$$\eta > 0 \Rightarrow \hat{Y} = 1, \quad \eta < 0 \Rightarrow \hat{Y} = 0.$$

This saves several keystrokes because $p > 0.5$ exactly when the log-odds are positive.

Expected check

The four values are

$$-1.6, \quad 1.6, \quad -0.75, \quad 0.55,$$

so the classifications are

$$0, \quad 1, \quad 0, \quad 1.$$

For interpretations, use the exponential key:

$$e^{0.06} = 1.06184, \quad e^{1.1} = 3.00417, \quad e^{-1.5} = 0.22313.$$

If an actual probability is required, evaluate

$$p = \frac{1}{1 + e^{-\eta}}.$$

4. Question 4: OLS proof

This question is mainly algebraic, so calculator use is limited. The key identities to remember are

$$(AB)^T = B^T A^T, \quad \text{Var}(AZ) = A \text{Var}(Z) A^T, \quad (X^T X)^T = X^T X.$$

The fx-580VN X can numerically check a small example using Matrix mode, but a numerical example is not a proof. In the exam, the marks come from the symbolic substitutions and matrix simplification.

Recommended exam-checking workflow

1. Write the statistical formula and method first.
2. Compute by hand far enough to show the examiner what you are doing.
3. Use the calculator to check means, matrix products, exponentials, and final decimals.
4. Keep at least 6–10 digits internally; round only in the final displayed answer.
5. For a classification threshold of 0.5, compare the logistic linear predictor with zero.

6. After a matrix calculation, check coefficient order carefully: intercept first, then slopes in the same order as the columns of X .